_briefing_combined

October 1, 2026

1 Week 4

Track-A Overture footprints in ten US downtown boxes, mapped to AlphaEarth 10 m cells, then pooled to 100 m tiles. Labels: `resi_def_v1`. Evaluation: leave-one-region-out (LORO).

Polygons are unstratified this week. Older Track-A draws capped each Overture class (~80/class) and ended ~27% residential. The unstratified draw keeps the natural mix (~90% residential among typed buildings, n=492,044). Training balance comes from undersampling residential *pixels*, not from oversampling rare non-residential polygons.

Head	Metric	Value
Logistic regression	macro-F1 (OOF)	0.806
Logistic regression	LORO mean ± std	0.779 ± 0.085
Ridge (3 fractions)	mean MAE	0.254
Ridge	MAE res / nres / empty	0.223 / 0.251 / 0.287

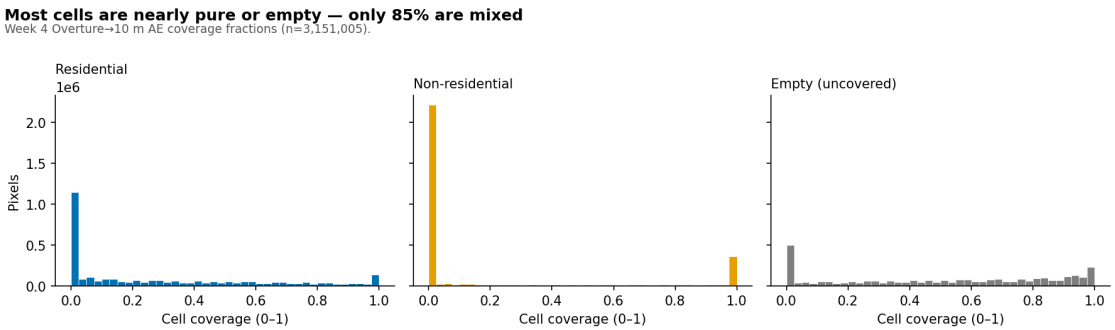
1.1 Data

1.1.1 Fraction ground truth at 10 m

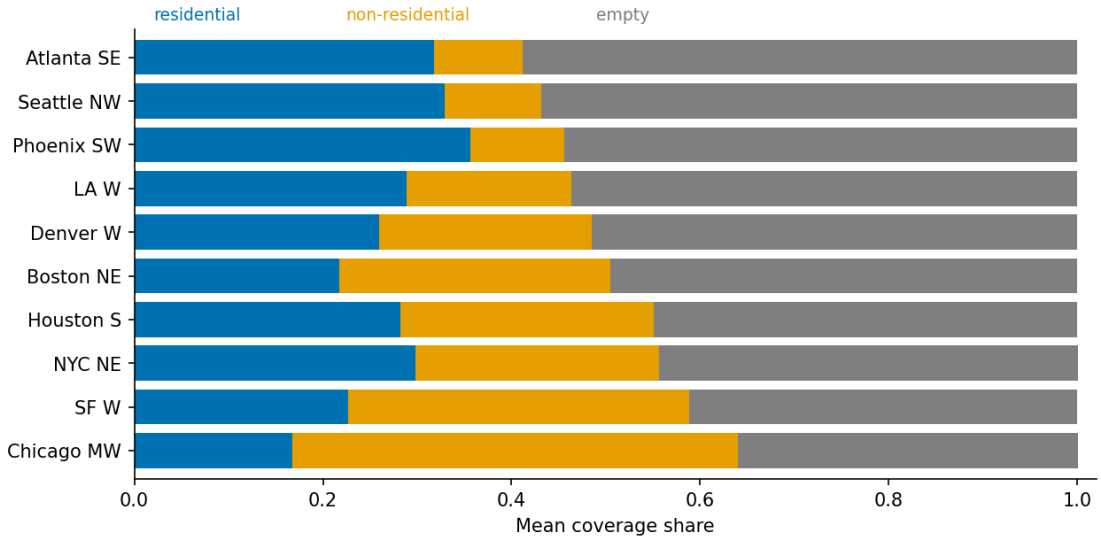
For each AlphaEarth cell that intersects at least one Track-A footprint:

```
frac_res   = area(union(RES footprints)  ∩ cell) / area(cell)
frac_nres  = area(union(NRES footprints) ∩ cell) / area(cell)
frac_empty = clip(1 − frac_res − frac_nres, 0, 1)
```

3,151,005 cells. Mean coverage: residential 0.288, non-residential 0.198, empty 0.515. Spec: docs/week4/pixel_eval_gt_design.md.



Empty share dominates every metro — buildings are a minority of AE cells

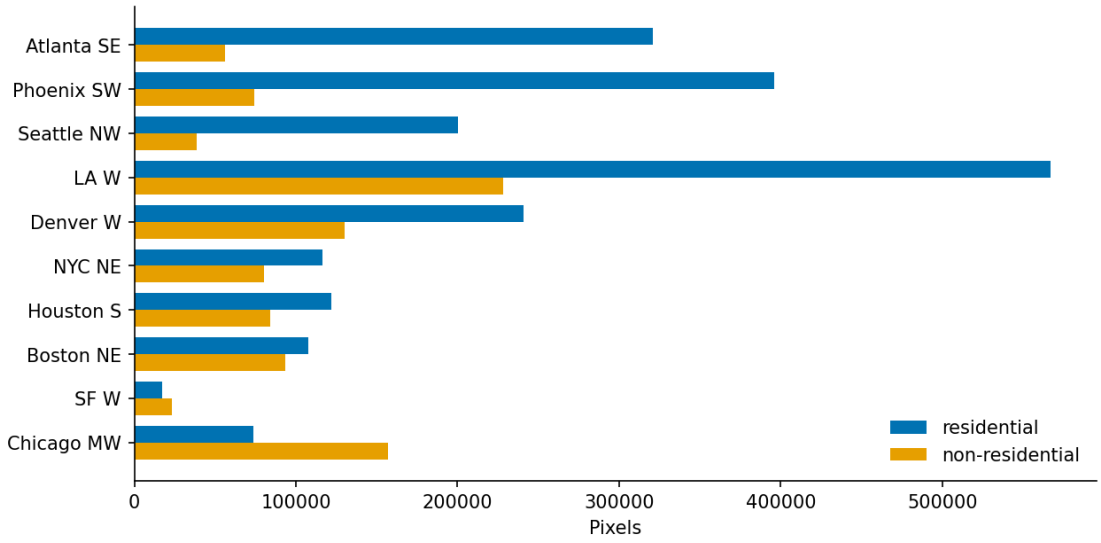


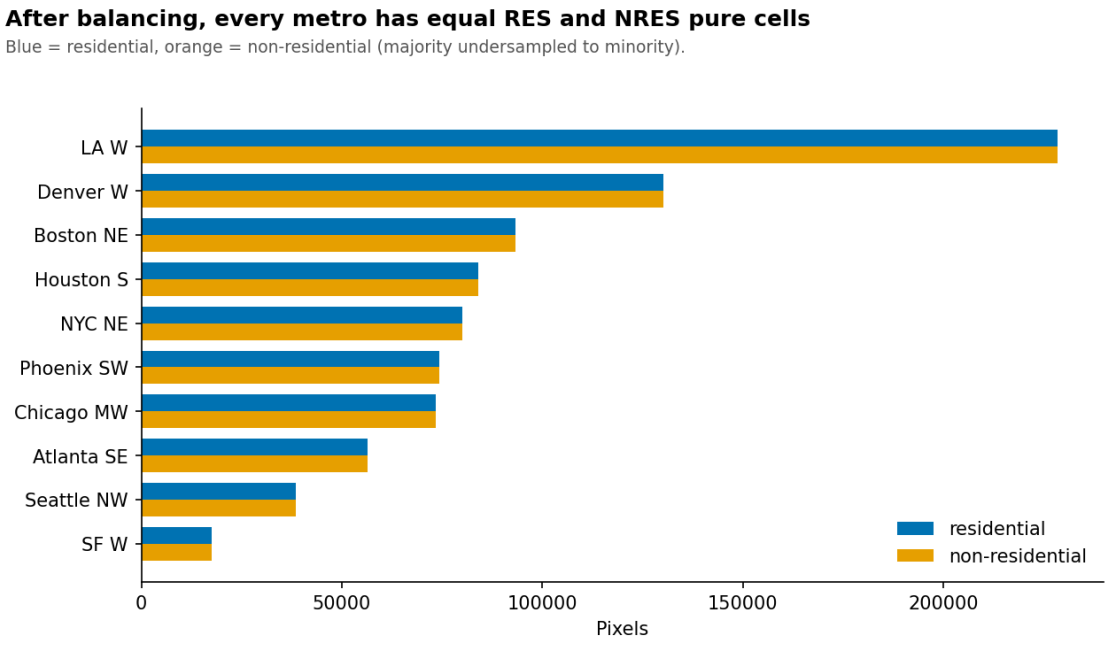
1.1.2 Pure-cell balance

A cell is pure residential if `frac_res > 0` and `frac_nres == 0` (swap for non-residential). Before balance: 2,162,534 RES / 965,515 NRES. Per metro, keep all NRES and draw RES down to match (seed 42) → 1,751,814 cells, 875,907 per class.

Before balancing, RES outnumbers NRES by ~1.7× (median metro)

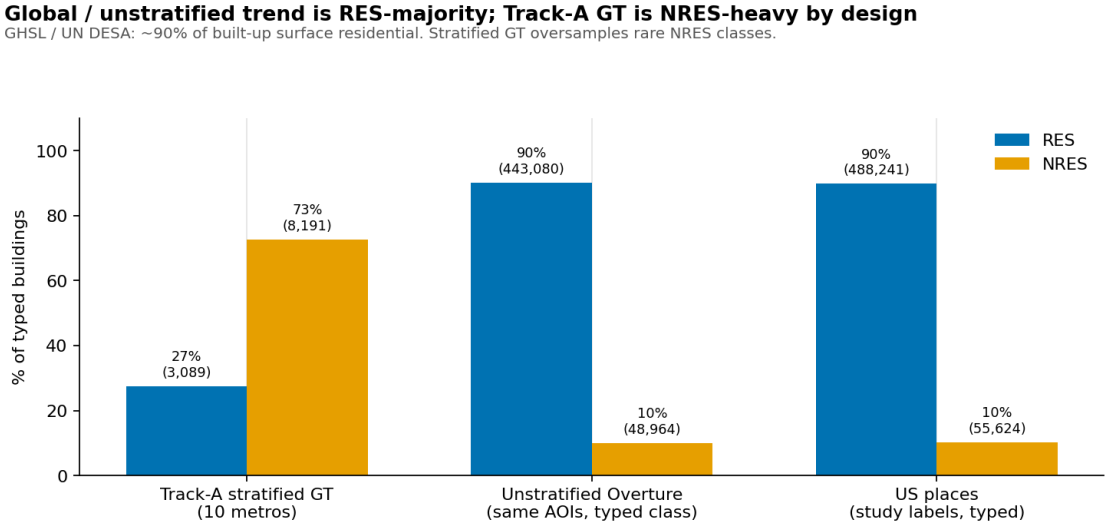
Pure cells only. Natural unstratified GT; sorted by RES:NRES ratio.





1.1.3 Why the old GT looked NRES-heavy

Stratified Track-A oversampled rare classes. Unstratified Overture in the same AOIs is ~90% residential by building count (~59% by typed area). Global GHSL / UN DESA built-up surface is also ~90% residential. Notebook: `06_res_nres_eda.ipynb`.

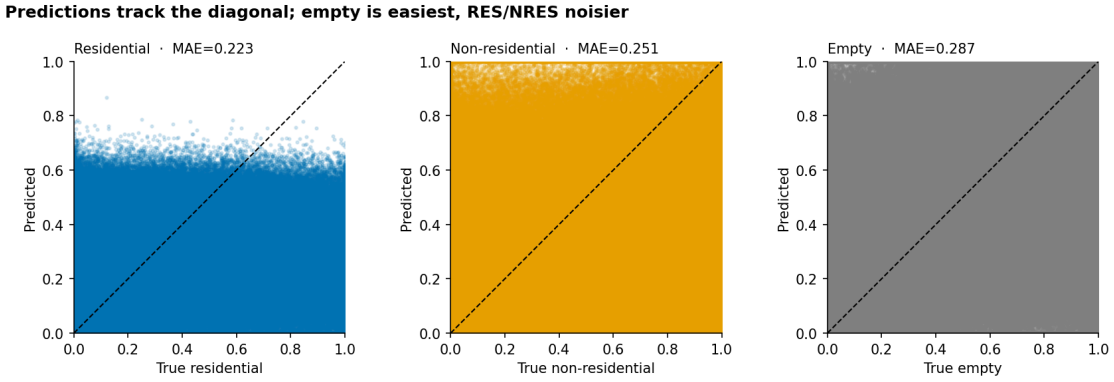


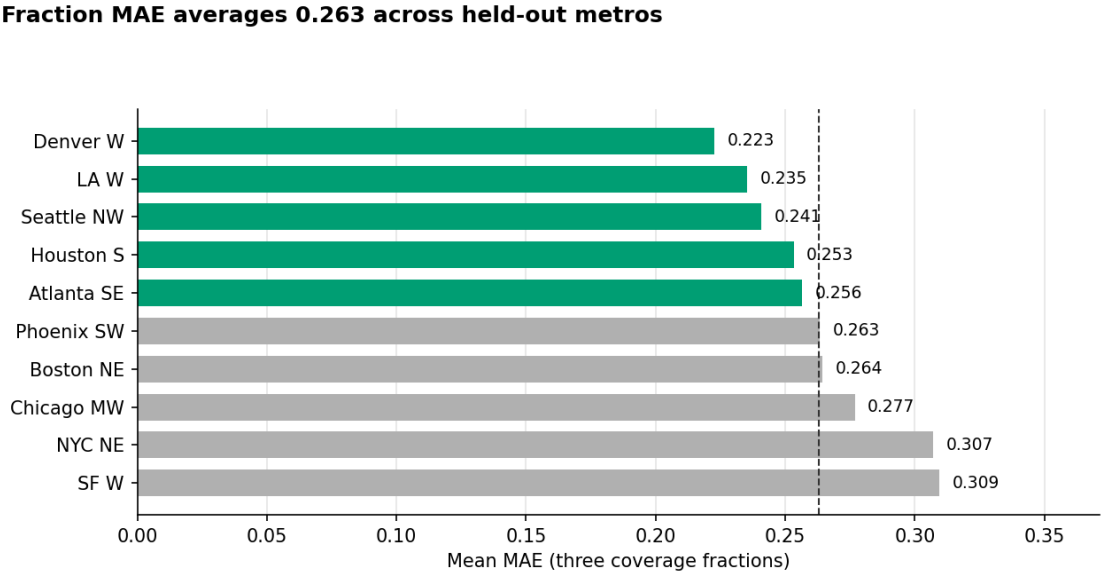
1.2 Models at 10 m

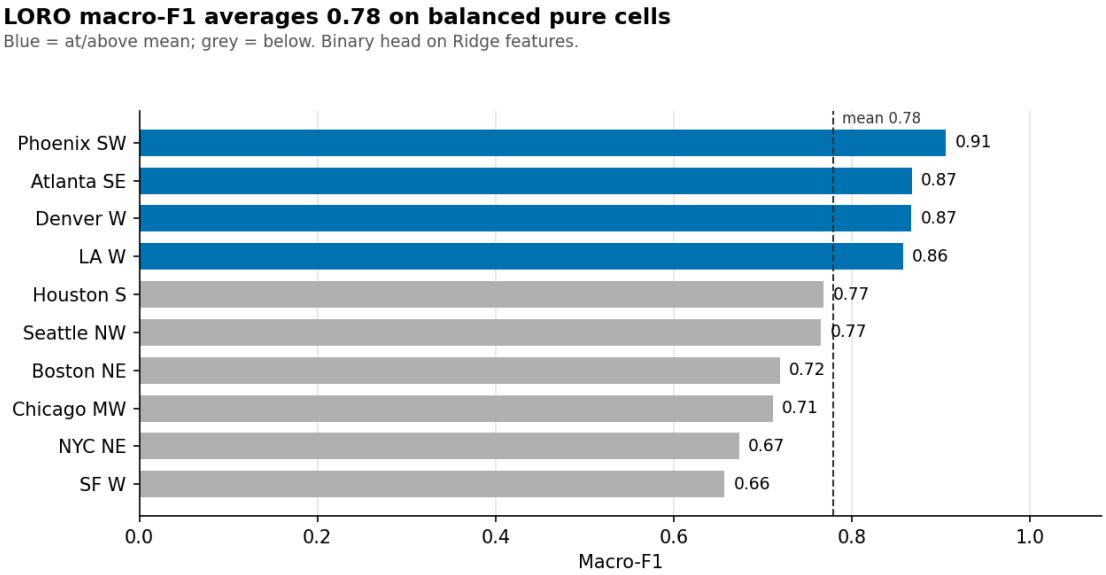
Same LORO folds on the balanced pure cells. Features: AlphaEarth `A00`–`A63`. No SMOTE.

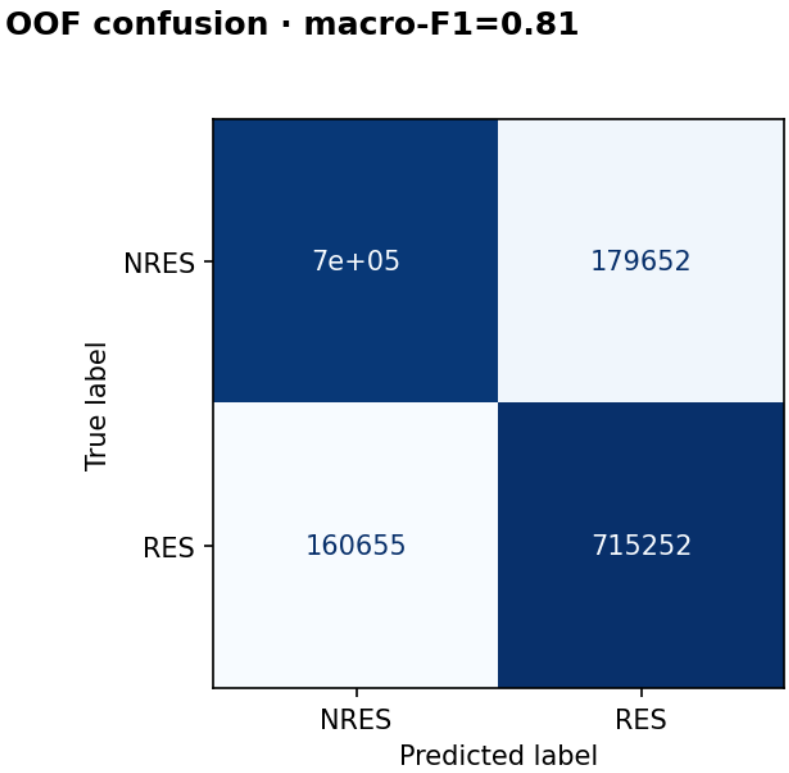
1. Ridge multi-output on `(frac_res, frac_nres, frac_empty)`.
2. Logistic regression on the binary stratum label.

Held-out fold models also score every pure cell in the full GT (3,128,049 cells; F1 0.820) so 100 m tiles have dense coverage.









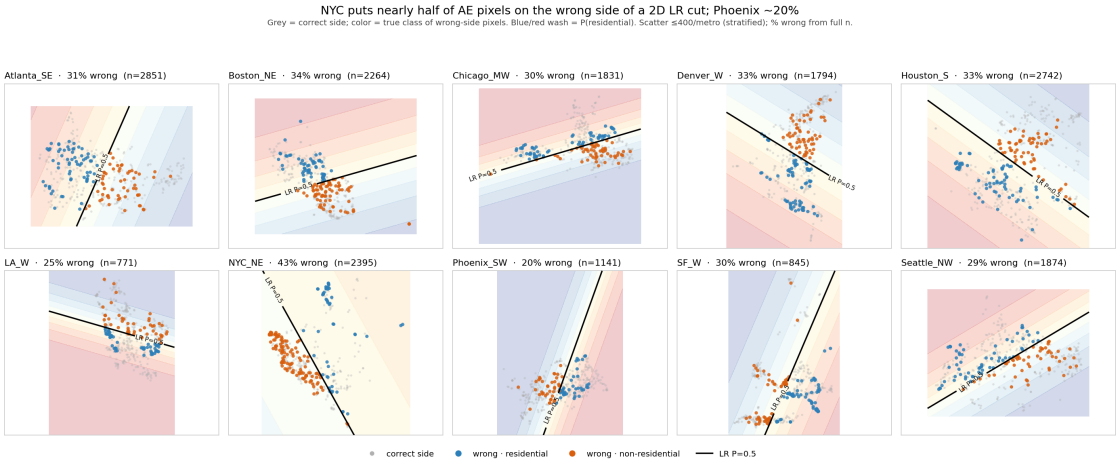
1.3 Region UMAP (Week 2 layout)

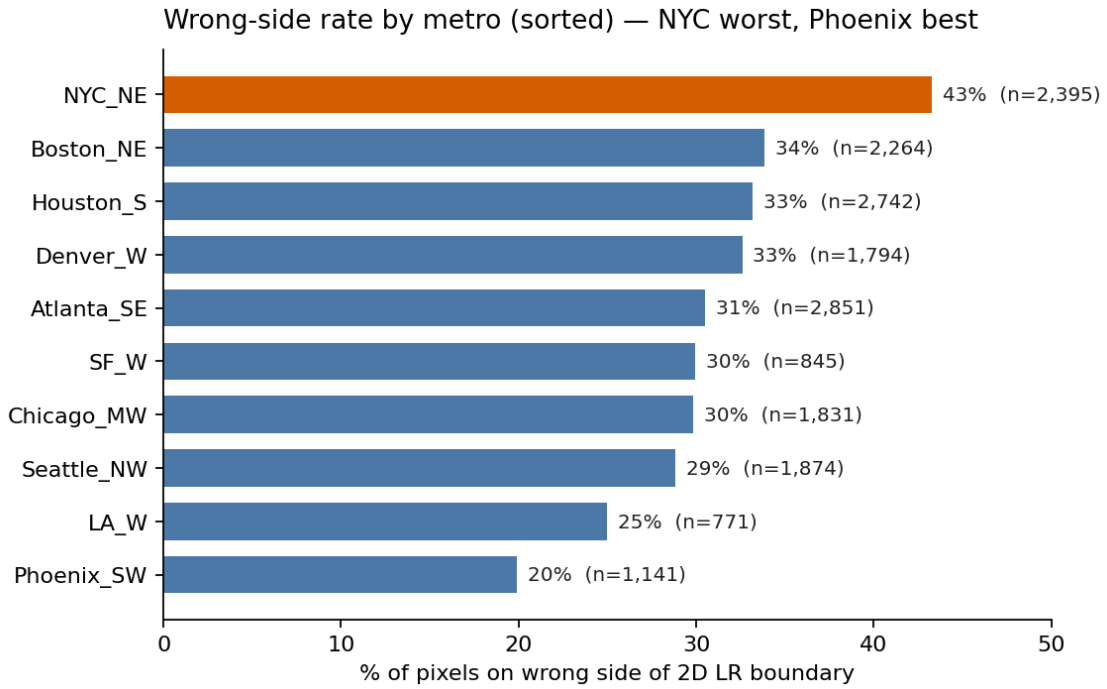
Same ten metros as the Week 2 CNN panel. Blue/red wash = 2D logistic `P(residential)`; black curve = `LR P=0.5`. Numbers below are *in-UMAP* linear separability, not 64-D LORO F1.

Week 2 = CNN on buildings. Week 4 = AlphaEarth on balanced pure 10 m cells (n=59,596). Mean in-UMAP F1: CNN 0.76, AE GT 0.69. Mean share of AE pixels on the wrong side of the cut: 31%

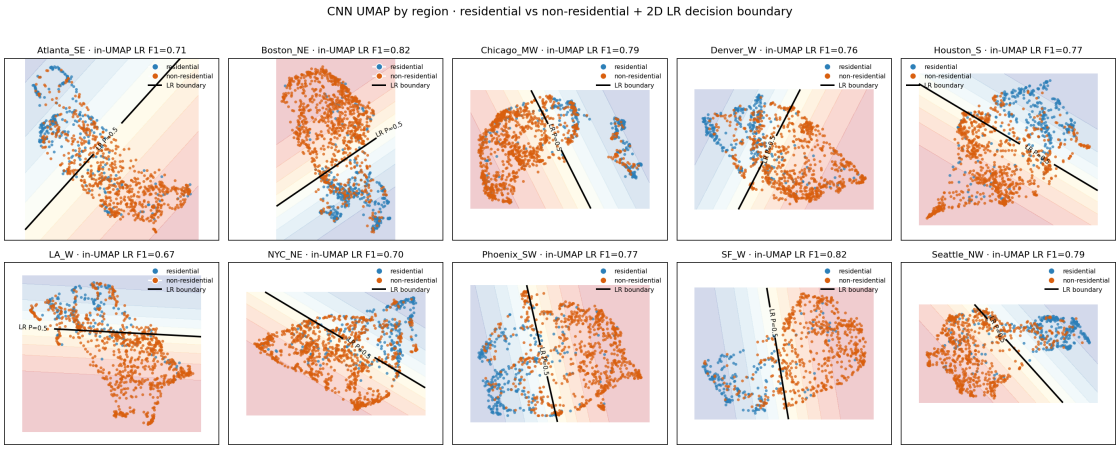
(NYC_NE 43%, Phoenix_SW 20%).

Grey = correct side; blue/orange = true class of wrong-side pixels. Panel titles print `% wrong` on the full metro; scatter is stratified to ≤400/metro.





Week 2 CNN reference (building patches, same panel order):

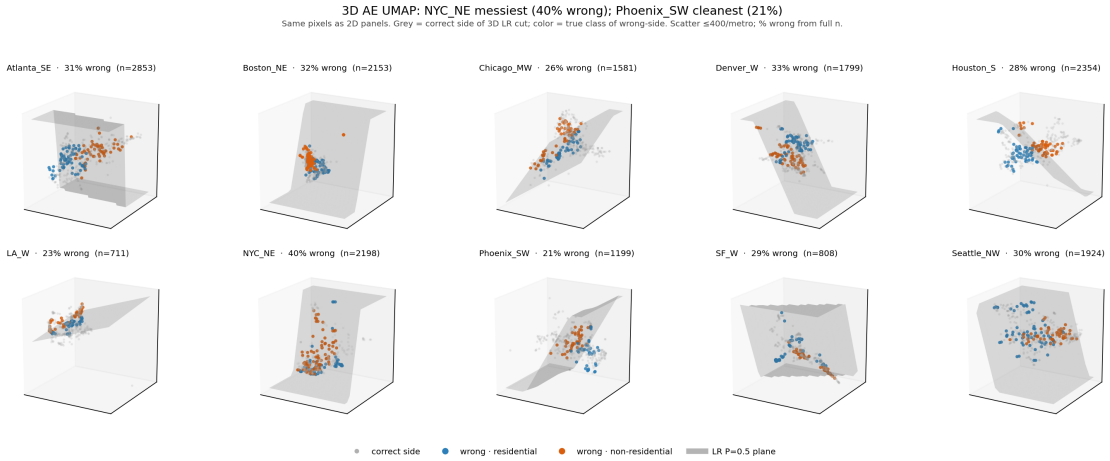


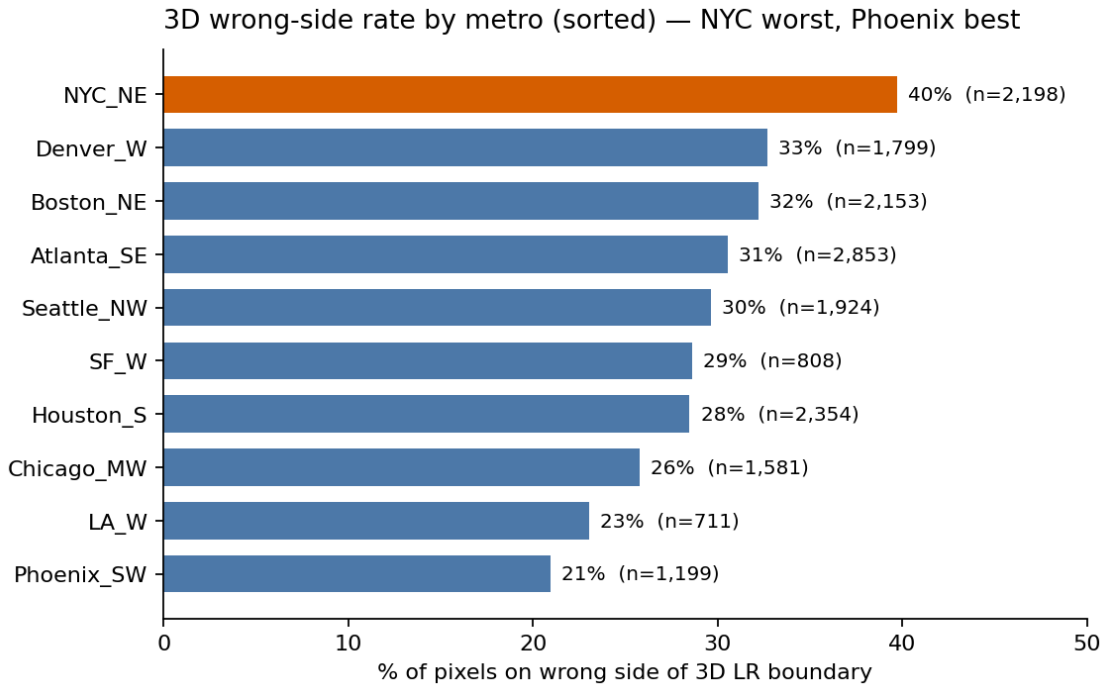
Region	W2 CNN F1	W4 AE GT F1	W4 % wrong
Atlanta_SE	0.71	0.69	31%
Boston_NE	0.82	0.66	34%
Chicago_MW	0.79	0.70	30%
Denver_W	0.76	0.67	33%
Houston_S	0.77	0.67	33%
LA_W	0.67	0.75	25%
NYC_NE	0.70	0.55	43%
Phoenix_SW	0.77	0.80	20%
SF_W	0.82	0.70	30%
Seattle_NW	0.79	0.71	29%
mean	0.76	0.69	31%

Dense GT-colored Week 4 panels (overplots; use wrong-side figure for counts): `outputs/week4_region_umap/figures/umap_gt_by_region_panels_residential_boundary.png`. Rebuild: `scripts/plot_week4_region_umap_boundary.py`.

1.4 Region UMAP 3D

Same 59,596 pixels, UMAP with 3 components, logistic cut is a plane. Grey = correct side; blue/orange = wrong-side by true class. Scatter ≤400/metro; `% wrong` from full n. Mean wrong-side 29% (NYC_NE 40%, Phoenix_SW 21%); mean in-UMAP F1 0.71. Interactive (rotate): `outputs/week4_region_umap_3d/figures/umap3d_gt_wrong_side_by_region.html`.





Region	2D % wrong	3D % wrong	2D F1	3D F1
Atlanta_SE	31%	31%	0.69	0.69
Boston_NE	34%	32%	0.66	0.68
Chicago_MW	30%	26%	0.70	0.74
Denver_W	33%	33%	0.67	0.67
Houston_S	33%	28%	0.67	0.72
LA_W	25%	23%	0.75	0.77
NYC_NE	43%	40%	0.55	0.60
Phoenix_SW	20%	21%	0.80	0.79
SF_W	30%	29%	0.70	0.71
Seattle_NW	29%	30%	0.71	0.70

Region	2D % wrong	3D % wrong	2D F1	3D F1
mean	31%	29%	0.69	0.71

Rebuild: `scripts/plot_week4_region_umap_3d.py`.

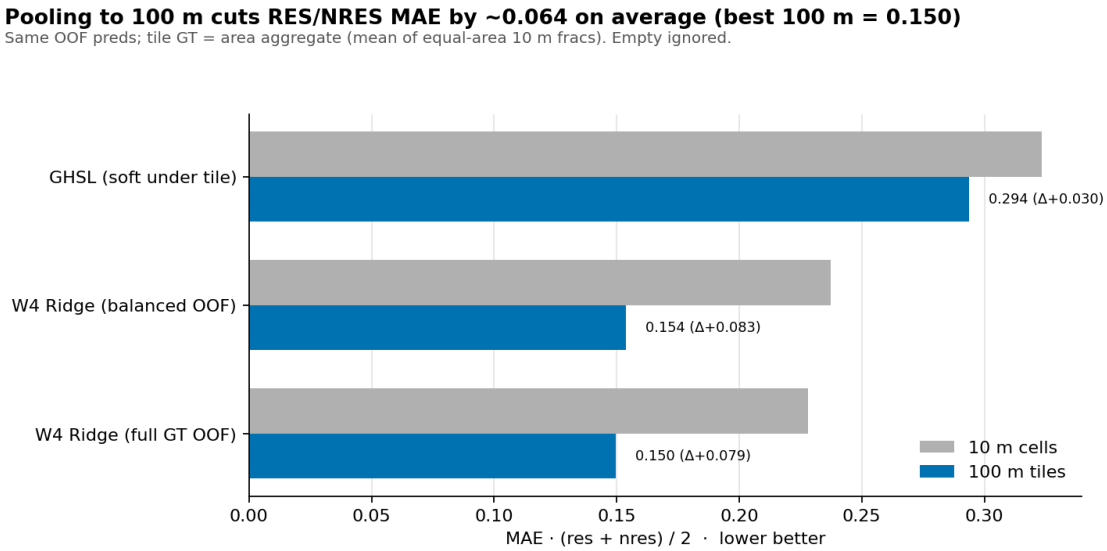
1.5 10 m cells vs 100 m tiles

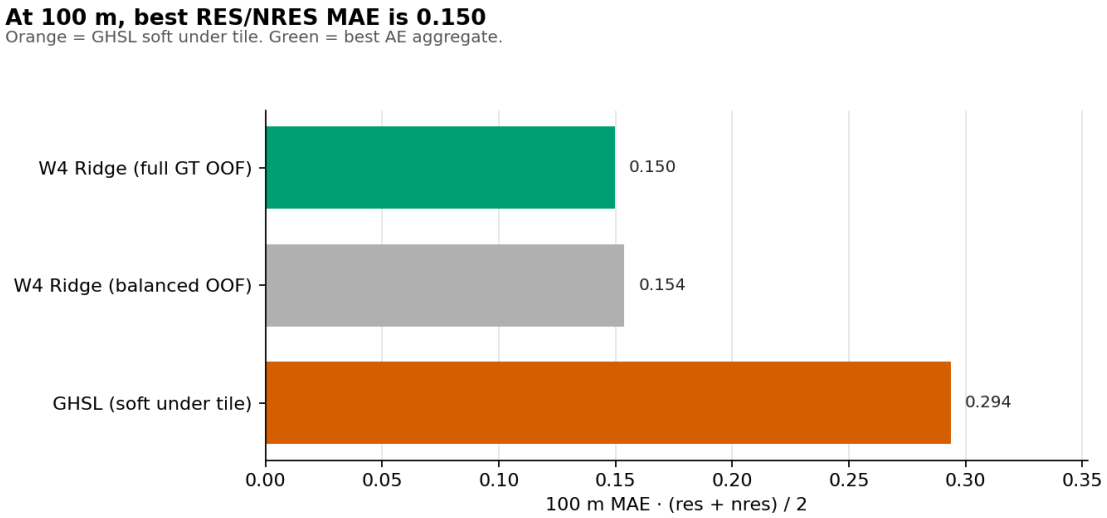
Drop tiles with fewer than five AE 10 m cells (`min` $\geq$ `5`); means and majority votes are unstable on thin tiles. Folders also keep `min` $\geq$ `20`. The three blocks below use different scoring sets; do not mix their numbers.

1.5.1 Fraction MAE (Ridge)

Full-GT OOF; mean of residential and non-residential MAE. Tile fractions = mean of equal-area 10 m cells.

	W4 Ridge	GHSL soft share
10 m	0.228	0.323
100 m (min $\geq$ 20)	0.150	0.294
Drop (10 m $-$ 100 m)	0.079	0.030

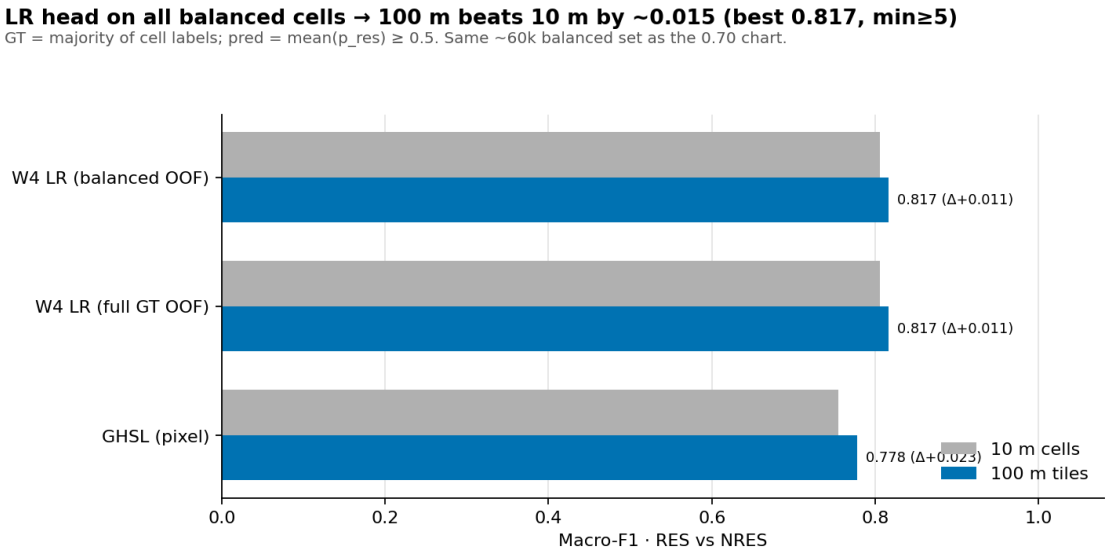




1.5.2 Binary F1 on tiles from balanced cells

Map each balanced train cell to its 100 m tile, keep tiles with ≥ 5 cells, predict with `mean(p_res)` $\geq$ `0.50`.

	F1 @ 10 m	F1 @ 100 m
W4 LR	0.806	0.817
GHSL	0.755	0.778

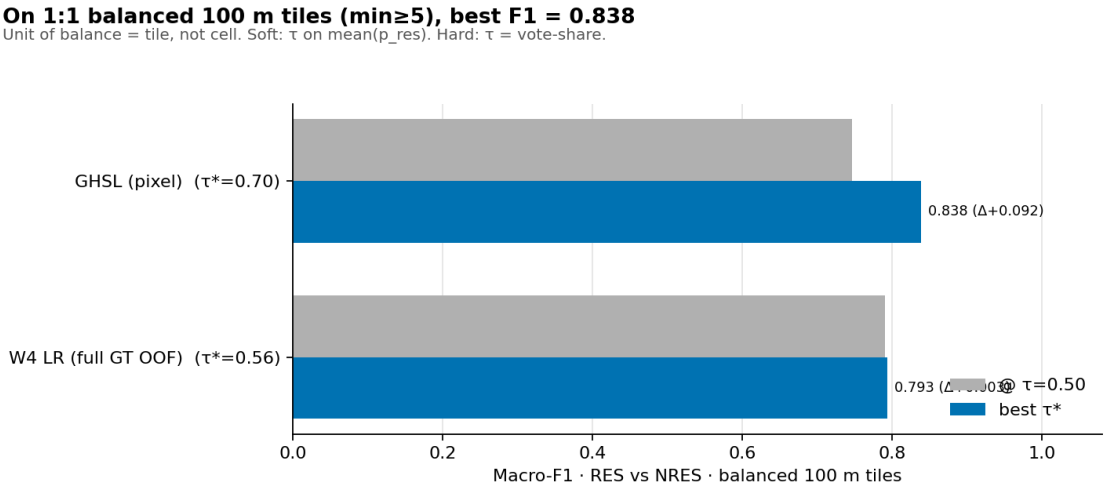


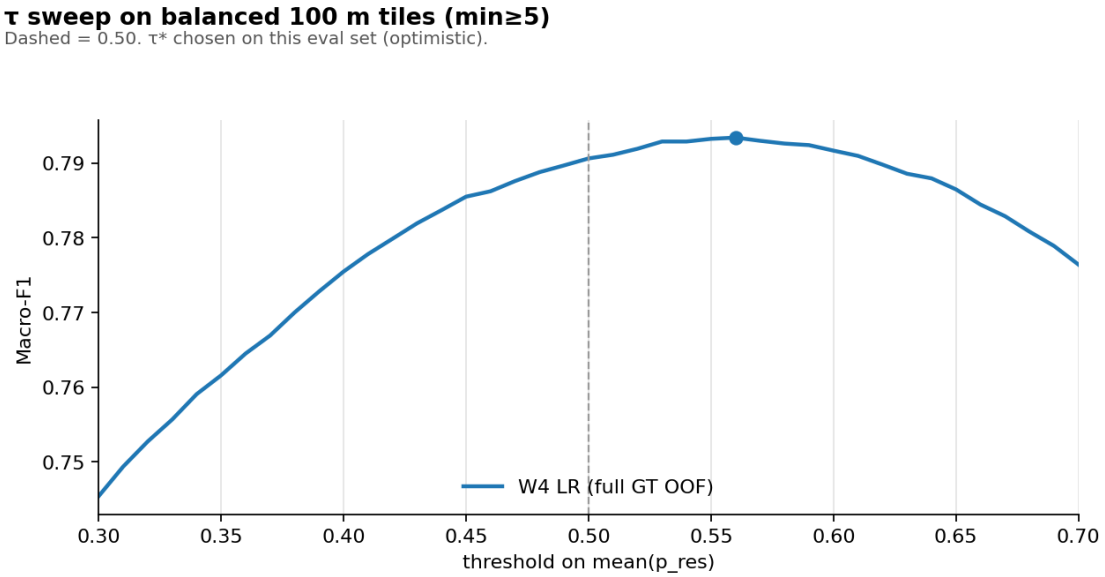
1.5.3 Binary F1 on a 1:1 tile sample

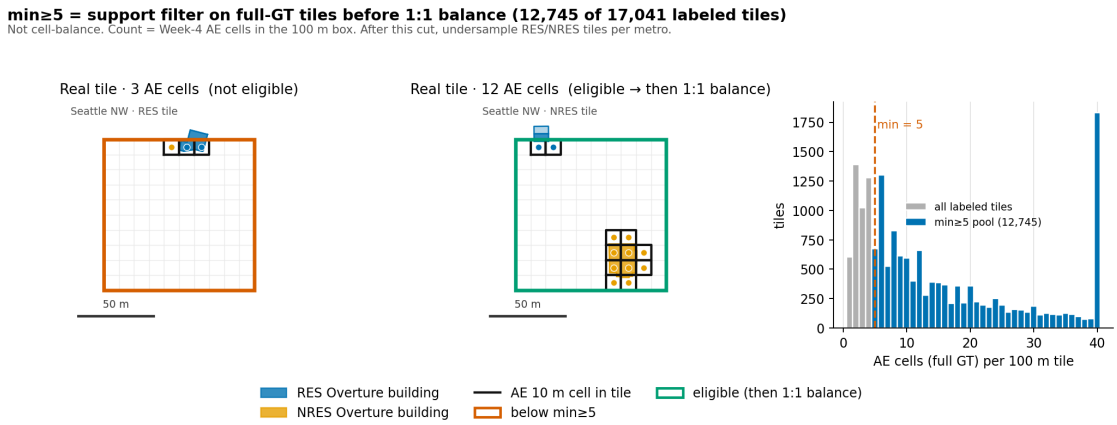
Label eligible tiles by `argmax(frac_res, frac_nres)`, then undersample per metro to equal RES/NRES (52,492 tiles at min ≥ 5). Score with `mean(p_res)` $\geq$ `0.50`.

	F1 @ τ=0.50	Best F1
W4 LR	0.791	0.793 (τ=0.56)
GHSL	0.746	0.838 (τ=0.70)

At the shared operating point τ=0.50, W4 LR beats GHSL. GHSL's best threshold is far from 0.50.







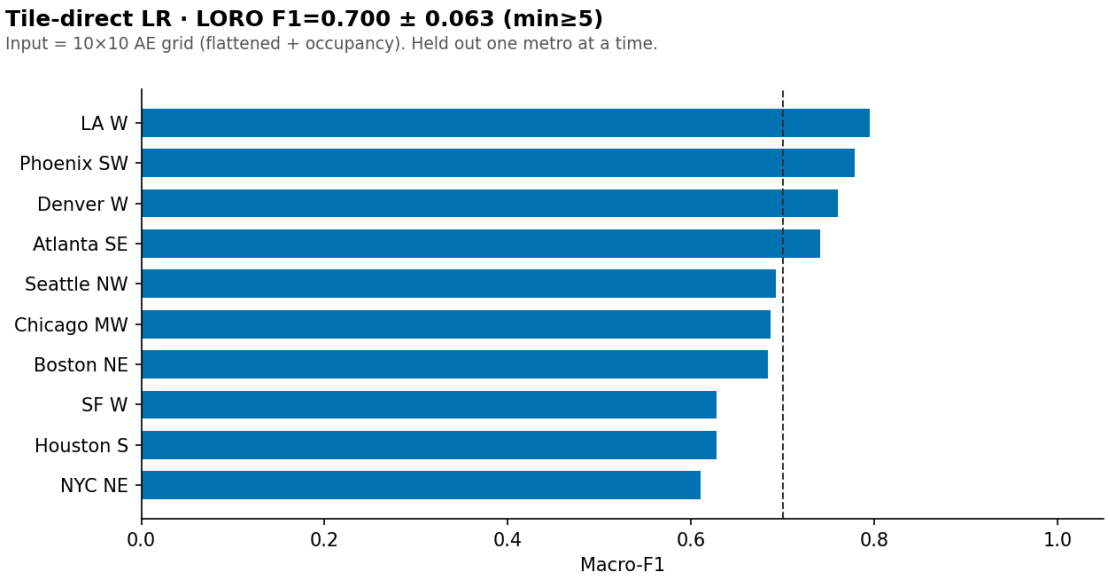
1.6 Tile-direct train (10×10 AE grid)

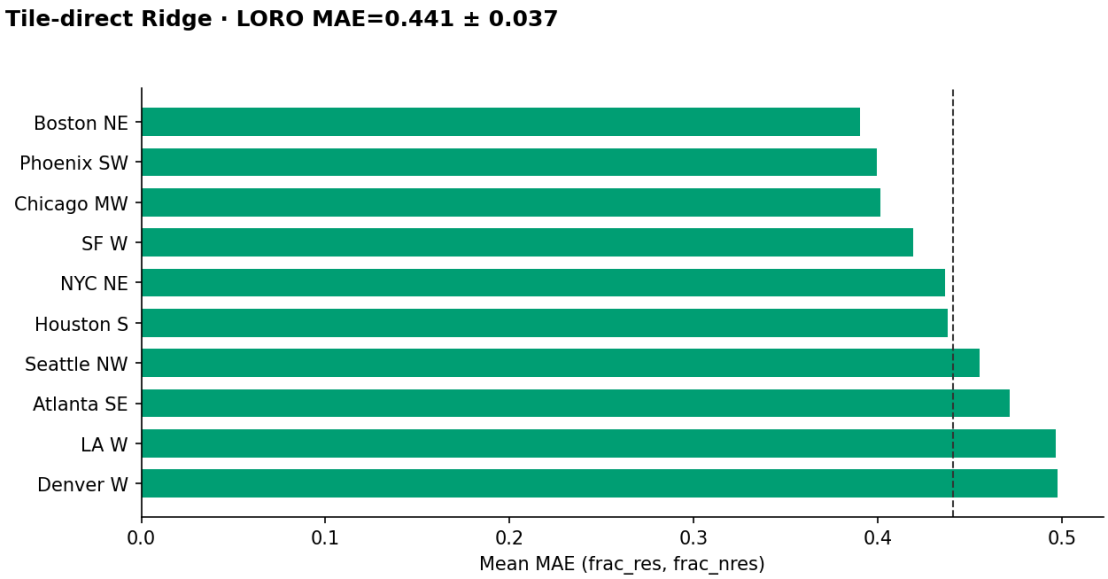
Train on the tile itself: flatten the 10×10 AlphaEarth grid (64-D per cell) plus occupancy mask → 6500-D. Same 1:1 balanced tiles as above. Heads: Ridge on `(frac_res, frac_nres)`, LR on RES vs NRES. LORO over metros. Notebook: `07_train_100m_tiles.ipynb`.

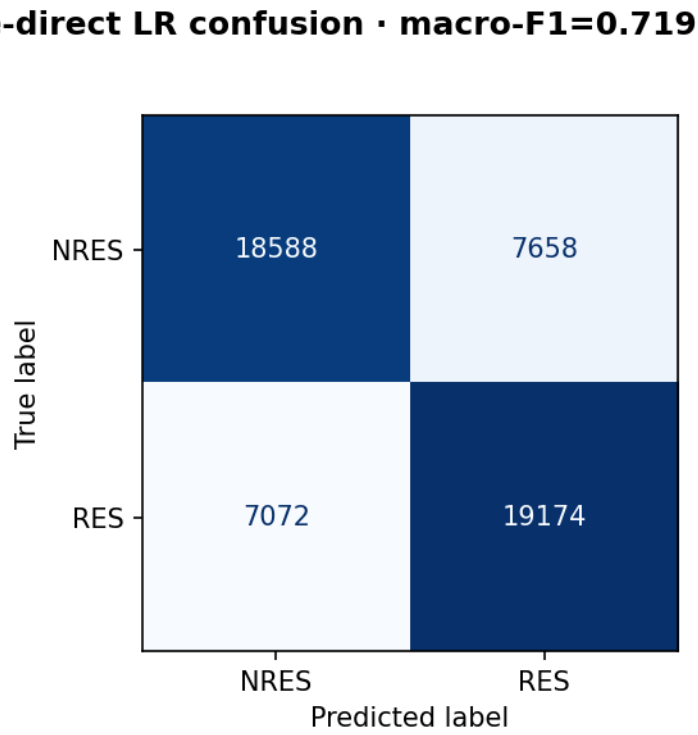
1.6.1 min ≥ 5

	Tile-direct	Cell-pool `mean(p_res)`≥0.50
OOF F1	0.719	0.791
LORO F1 mean ± std	0.700 ± 0.063	—
Mean MAE (frac_res, frac_nres)	0.448	—
Δ F1 (direct − pool)	-0.071	

Linear heads on the flattened grid trail cell-pool here.

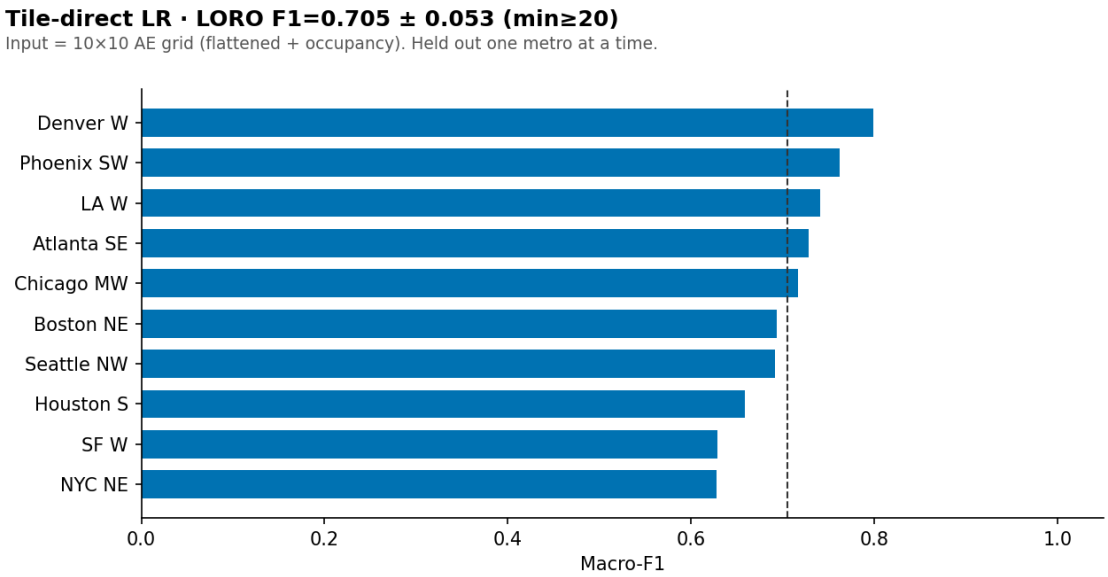






1.6.2 min ≥ 20

	Tile-direct	Cell-pool @ 0.50
OOF F1	0.726	0.808
Mean MAE	0.401	—
Δ F1	-0.082	



1.7 Paths

Artifact	Path
Fraction GT	`datasets/week4_pixel_gt/`
Balanced pure cells	`datasets/week4_pixel_gt_balanced/`
Train OOF + checkpoints	`outputs/week4_pixel_balanced_train/`
100 m fraction / LR eval	`outputs/week4_100m_fraction_eval/`
1:1 tile eval	`outputs/week4_100m_tiles_balanced/`
Tile-direct train	`outputs/week4_100m_tile_train/`
RES/NRES EDA	`outputs/week4_res_nres_eda/`
Region UMAP	`outputs/week4_region_umap/`
Region UMAP 3D	`outputs/week4_region_umap_3d/`

2 Week 4 follow-up

This note starts after `00_briefing.ipynb`. Same ten downtown boxes, same `resi_def_v1` labels, same leave-one-metro-out split. The question here is whether a small embedding of a 100 m tile separates residential from non-residential inside each metro.

Two summaries. Linear models on AlphaEarth already sit near 0.76 LORO macro-F1 at 100 m, and PLS does not beat the 64-D logistic regression. Forcing the ViT down to 3 numbers raised the held-out gap and left F1 about where v1 and v3 were. Atlanta, Phoenix, and Denver clear a held-out gap of 3. New York does not.

Run	LORO macro-F1	Mean held-out gap
100 m mean-pool logistic regression	0.761	—
ViT, 128-D class token	0.756	—
v1, BCE + SupCon	0.732	1.76

Run	LORO macro-F1	Mean held-out gap
v2, detached head	0.632	2.12
v3, same loss, longer training	0.735	1.71
v4, within-region pull and gap	0.738	2.13

Gap = distance between the two class centers ÷ mean radius of each class. A gap of 3 was the training target. A gap near 1 is one cloud. Raw sep is not comparable across runs: v4's clouds are much smaller than v1's, and the ratio is the number to compare. Each fold has its own 3D coordinates.

2.1 Linear baselines

100 m features are the mean of the AlphaEarth cells inside the tile. Tiles need at least 5 occupied cells, then residential and non-residential are balanced (n=52,492). The 10 m numbers use the balanced pure cells (n=1,751,814).

Scale	Model	Width	LORO macro-F1
10 m cells	64-D logistic regression	64	0.779 ± 0.085
10 m cells	PLS-DA + logistic regression	32	0.775 ± 0.088
10 m cells	PCA + logistic regression	32	0.759 ± 0.084
10 m cells	L1 on 32 PLS scores	32, C=30	0.751 ± 0.105
10 m cells	t1 only	1	0.735 ± 0.117
100 m mean pool	64-D logistic regression	64	0.761 ± 0.092
100 m mean pool	PLS-DA + logistic regression	24	0.762 ± 0.096
100 m mean pool	PCA + logistic regression	32	0.749 ± 0.090
100 m mean pool	L1 on 32 PLS scores	32, C=30	0.738 ± 0.098
100 m mean pool	t1 only	1	0.720 ± 0.119
100 m	flat 10×10 grid, logistic regression	6500	0.700
100 m	ViT, 128-D class token	128	0.756 ± 0.119

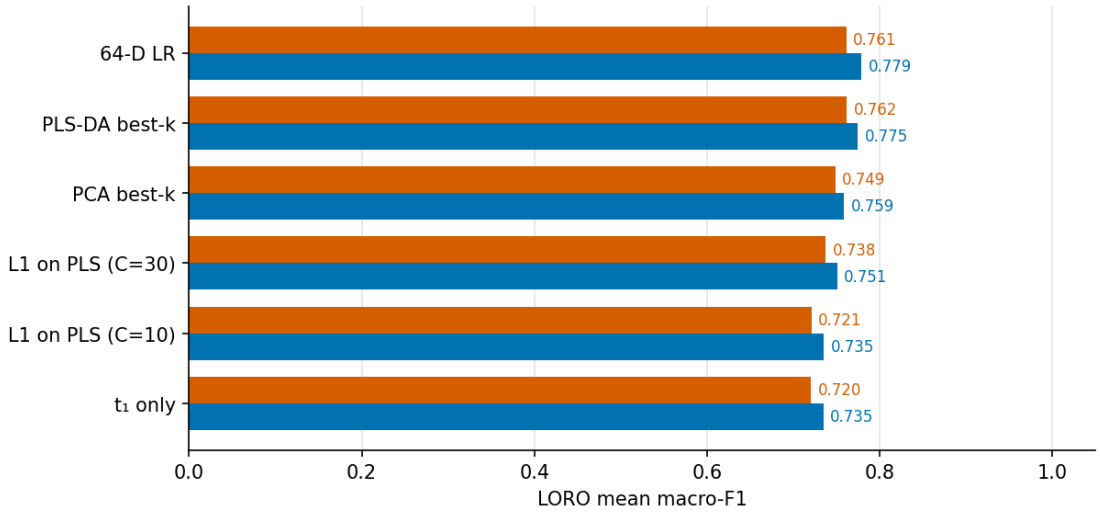
Moving from 10 m cells to a 100 m mean costs 0.018 F1 on the 64-D logistic regression (0.779 → 0.761) and 0.013 on the best PLS-DA head. PLS-DA at 100 m, k=24, is 0.762, level with the full 64-D head. PCA is lower. The first PLS score alone is 0.720.

L1 on the 32 PLS scores drops the mean number of nonzero weights from 32 to about 1.2 at C=10, and that survivor is t1 on every fold where anything survives (t1 survival 1.0, t2 survival 0.2). LORO F1 at that setting is 0.735, the same neighborhood as the t1-only head. Stronger L1 zeros every score.

k=9 is the smallest PLS width within 0.02 F1 of the 64-D logistic regression (F1 0.761, gap 0.018). The ViT reads those 9 scores. Empty cells in the 10×10 grid are masked.

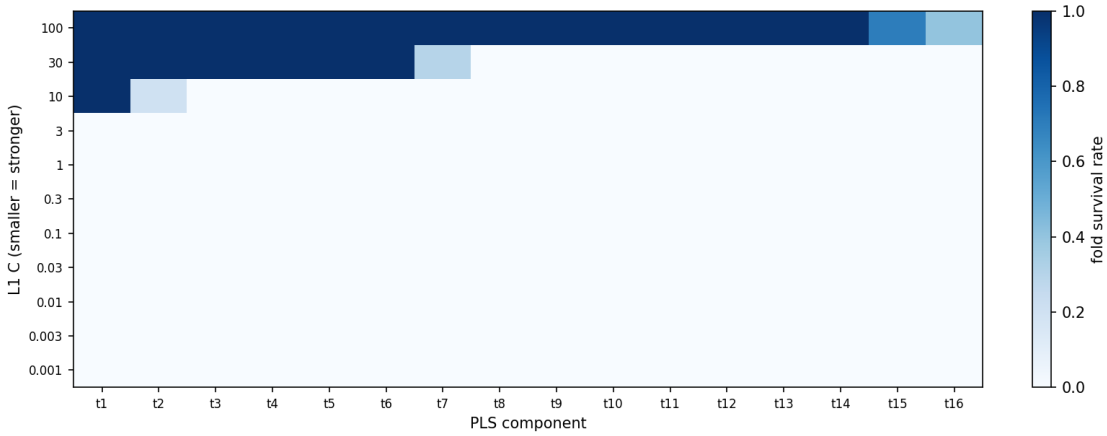
Every head is weaker at 100 m; t_1 still carries most of the signal

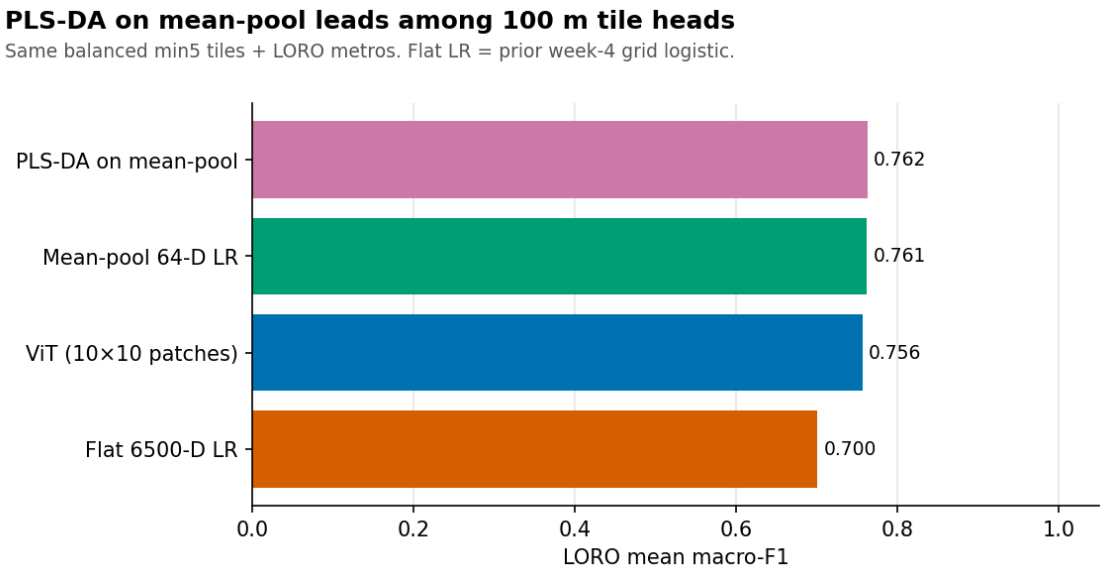
Blue = 10 m pure cells. Orange = 100 m mean-pooled AE. Same LORO metros.



L1 on PLS scores: at strong penalty only t1 (and sometimes t2) survive

Cell = fraction of LORO folds with |coef| > 1e-6. Strong L1 = bottom rows.





Notebooks: 08_pls_comparison.ipynb, 09_l1_pls_sparsity.ipynb, 10_pls_10m_vs_100m.ipynb.

2.2 Why the 3D runs changed

The 128-D ViT matches the mean-pool logistic regression and does not show two clusters, because there is no 3D space to look at. The next runs keep the same ViT (d_model 128, 4 layers, 4 heads) and replace the class token with 3 numbers. The aim is two residential / non-residential clusters inside each metro after fine-tuning.

v1 adds supervised contrastive loss on the L2-normalized 3D vector, λ=1, temperature 0.1, and lets the classifier gradient into the encoder. Held-out F1 is 0.732. The clouds are horseshoes: the linear head only needs one direction, so the extra two axes collapse into a curve. The tips of the curve are class-pure. The bottom is mixed. SupCon's minimum for two classes is two opposite poles. This run did not reach that minimum.

v2 makes four changes: stop the classifier gradient at the 3D vector, use a cosine head with a learnable scale, set λ=10, and anneal temperature from 0.5 to 0.1. The encoder's only training signal is SupCon. Held-out F1 falls to 0.632. SupCon stays near the chance value for a batch of 256 (about $\log 255 \approx 5.54$). Atlanta is the clear failure: the checkpoint is epoch 1, training stops at epoch 8, and every held-out tile is predicted non-residential (F1 0.333). v1 Atlanta was 0.860. The stop rule was validation F1, while training BCE was still falling.

v3 keeps the v2 loss and changes the schedule. Every fold runs at least 35 epochs, then stops on validation loss (BCE + λ τ SupCon), patience 6, cap 60. F1 is logged and does not choose the checkpoint. Held-out F1 returns to 0.735, in line with v1. All 10 folds run to the cap. Temperature is still annealing across the 60-epoch budget, so the weighted validation loss keeps falling while validation SupCon moves back toward chance. Longer training did not make two poles.

v4 drops SupCon, the detach, and the temperature. Each batch is one metro, half residential and half non-residential. On the raw 3D vectors:

```
loss = BCE(scale * (z · unit w), y)
     + mean squared distance to the class mean
     + relu(3 − sep/within)^2
```

Both the classifier and the cluster terms send gradient into z. The stop metric is the mean sep/within on a 10% split of the other metros. Higher is better. The floor is still 35 epochs, patience 6, cap 60. F1 does not choose the checkpoint. Held-out F1 is 0.738. Mean held-out gap is 2.13, compared with 1.71 in v3.

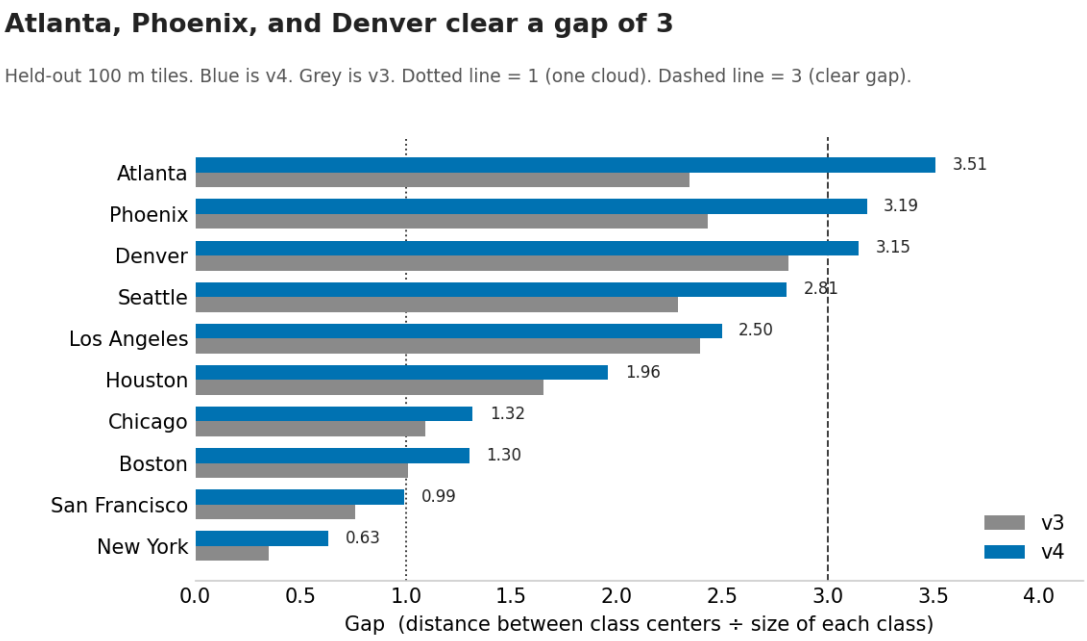
Notebooks: 12_vit_100m_3d.ipynb through 15_vit_100m_3d_v4.ipynb.

2.3 Held-out metros under v4

Atlanta, Phoenix, and Denver are the held-out metros with gap ≥ 3. The gap is higher than v3 in every metro. F1 moves by metro and the mean is flat (0.738 vs v3 0.735). Chicago drops from 0.622 to 0.553. New York stays one cloud: gap 0.63, F1 0.586.

The validation column is the mean gap on a 10% split of the other metros. That is the number training stopped on. New York's fold reached a validation gap of 3.33 and a held-out gap of 0.63.

Metro	v4 F1	v3 F1	v1 F1	v4 gap	v3 gap	v4 val gap	epochs
Atlanta	0.898	0.840	0.860	3.51	2.34	2.69	35
Phoenix	0.878	0.849	0.888	3.19	2.43	2.75	40
Denver	0.862	0.852	0.867	3.15	2.81	2.85	42
Seattle	0.824	0.787	0.788	2.81	2.29	2.95	45
Los Angeles	0.793	0.842	0.786	2.50	2.39	3.00	35
Houston	0.709	0.711	0.693	1.96	1.65	2.94	41
Chicago	0.553	0.622	0.610	1.32	1.09	3.07	35
Boston	0.693	0.698	0.702	1.30	1.01	3.13	45
San Francisco	0.584	0.595	0.573	0.99	0.76	3.06	35
New York	0.586	0.552	0.554	0.63	0.35	3.33	35



2.4 How to read a cluster figure

Left panel: the three encoder outputs, unrotated. The gray plane is the classifier cut, $z \cdot w = 0$. Right panel: the same tiles on the classifier score and the widest direction perpendicular to it. The dashed line is the same cut. Residential is on the positive side of the score.

Color is the mix of the two class centers. Orange is non-residential, blue is residential, mauve is where both centers still reach. Influence of a center falls with distance, and the radius reaches at least halfway to the other center, so the overlap is a blend instead of one class drawn on top of the other. Marker size falls off with that influence. The white-edged dots are the two centers.

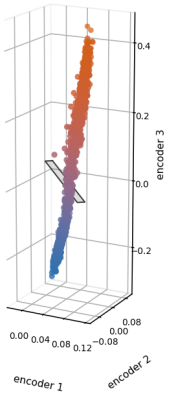
A high gap does not mean an empty band between the classes. Denver's held-out gap is 3.15 and its F1 is 0.862. 89% of non-residential tiles have score below 0, and 83% of residential tiles have score above 0. Along the score, the leftmost 30% of tiles are 9% residential, the middle 40% are 46% residential, and the rightmost 30% are 96% residential. Almost every tile of either class still sits inside the other class's score range (over 99.9% of each). The cut sorts the mass. The supports overlap.

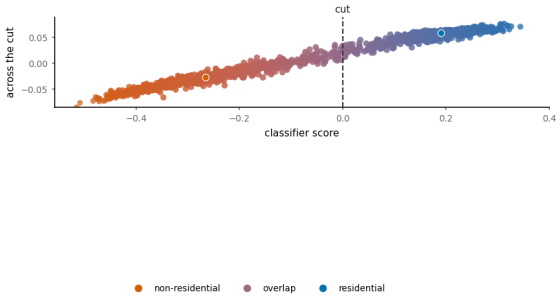
The geometry is a thin line along the class axis. A different camera angle does not open a gap.

Atlanta clears the gap. Denver is the high-F1 case above. New York is the held-out cloud.

Atlanta: residential and non-residential are apart

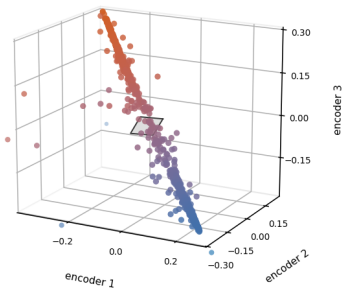
Held-out tiles. F1 0.898. Gap 3.51. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.

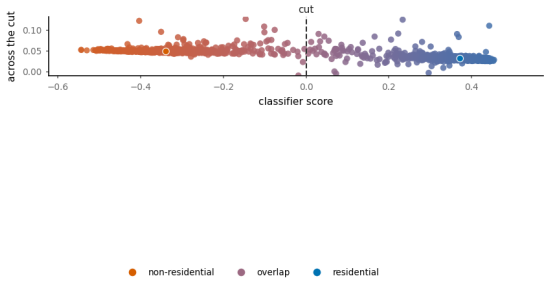




Denver: residential and non-residential are apart

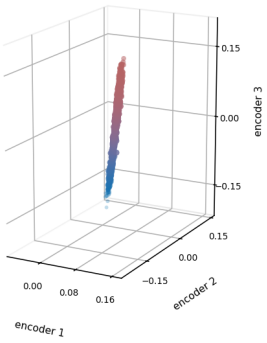
Held-out tiles. F1 0.862. Gap 3.15. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.

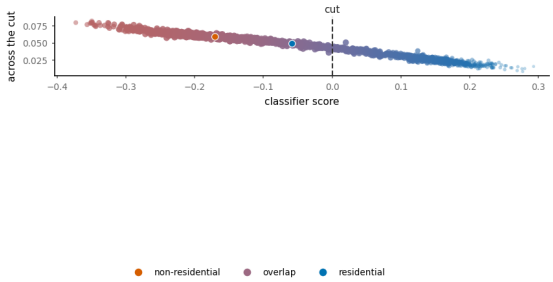




New York: residential and non-residential are one cloud

Held-out tiles. F1 0.586. Gap 0.63. Mauve is where both centers still reach. Influence falls with distance from a center. The plane and the dashed line are the classifier cut.

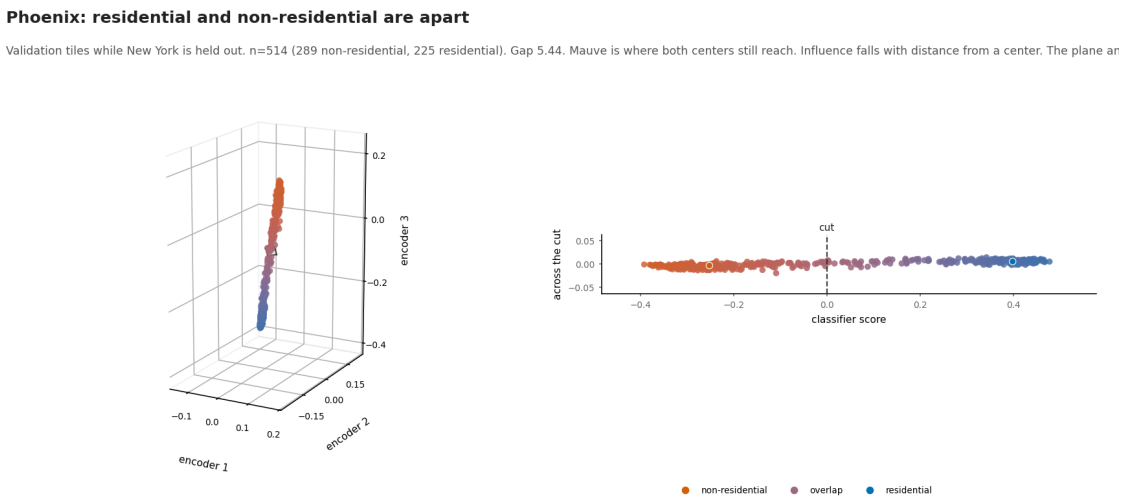


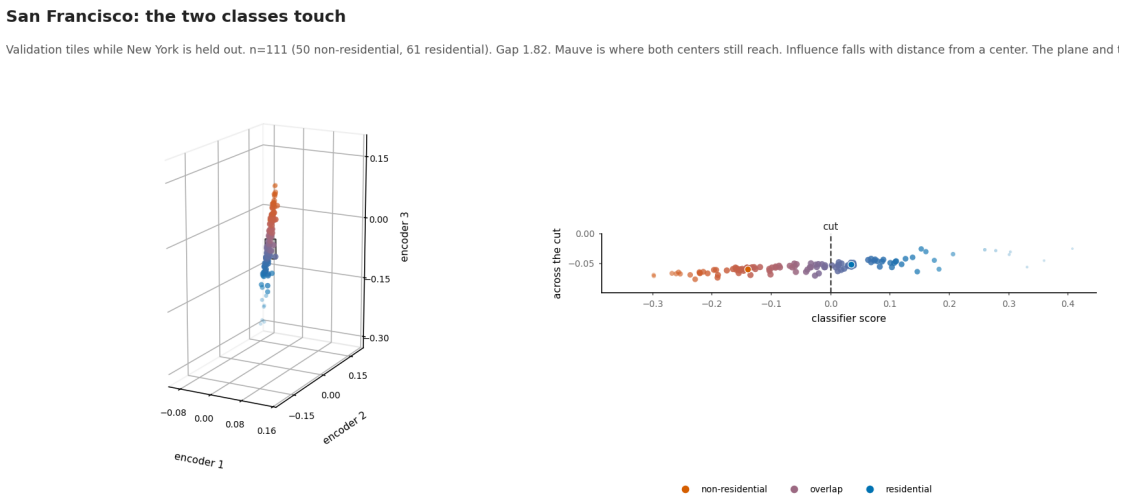


2.5 Validation while New York is held out

The New York fold's best validation gap is 3.33, measured on 10% of the other metros at the checkpoint (logged best 3.334). The same checkpoint's held-out New York gap is 0.63. The other cities in that fold do separate. They are still thin lines with the classes at opposite ends.

Metro in the New York fold's validation split	Gap	Sep	Within	n
Phoenix	5.44	0.553	0.102	514
Atlanta	4.98	0.458	0.092	367
Denver	3.74	0.468	0.125	743
Los Angeles	3.71	0.440	0.118	1068
Seattle	3.59	0.396	0.110	225
Houston	2.80	0.324	0.116	637
Chicago	2.01	0.264	0.131	505
Boston	1.90	0.261	0.138	550
San Francisco	1.82	0.173	0.095	111





The other held-out figures, and the rest of this validation split, are in `08_pls_comparison.ipynb`.

2.6 Paths

Artifact	Path
PLS LORO	`outputs/week4_pls_loro/`
10 m vs 100 m	`outputs/week4_pls_10m_vs_100m/`
L1 on PLS scores	`outputs/week4_l1_pls_loro/`
ViT 128-D	`outputs/week4_vit_100m/`
3D v1–v3	`outputs/week4_vit_100m_3d/`, `_v2/`, `_v3/`
3D v4 folds, checkpoints, figures	`outputs/week4_vit_100m_3d_v4/`
This note	`notebooks/week4/16_briefing.ipynb`